EJS (Embedded JavaScript Templates)

HTML + small pieces of JavaScript to display dynamic data

EJS (Embedded JavaScript Templates) is a server-side templating engine for Express that lets you generate HTML using JavaScript.

Why EJS?

Express does not render HTML by itself.

EJS provides:

- HTML templates
- Variables
- Loops & conditionals
- Partials (reusable layouts)

Similar to:

- Blade (PHP/Laravel)
- Jinja (Python/Flask)

Express + EJS Flow

```
Browser → Express Route → EJS Template → Final HTML
```

Example:

```
app.get('/hello', (req, res) => {
  res.render('hello', { name: 'Sam' });
});
```

Express renders `views/hello.ejs`, and replaces name with 'Sam'.

EJS Example Template

We can use JavaScript expressions in an EJS template inside `<%= ... %>`.

`hello.ejs`:

```
<h1>Hello <%= name %>!</h1>
```

Renders:

```
<h1>Hello Sam!</h1>
```

Insert variable

```
<p><%= user.email %></p>
```

Loop

```
<% posts.forEach(p => { %>
  <li><%= p.title %></li>
<% }) %>
```

If

```
<% if(isAdmin){ %>
  <button>Admin</button>
<% } %>
```

Enabling EJS in Express

```
app.set('view engine', 'ejs');
```

Express automatically uses `views/` as the folder.

Partials: `include`

A partial is a reusable piece of a template. The pages of our example include the navigation bar `nav.ejs` with:

```
<%- include('nav.ejs') %>
```

- `<%= ... %>` **escapes** the value (safe for user data).
- `<%- ... %>` inserts the value **without escaping** (needed to insert HTML, such as an included file).

Example: index_ejs.js

```
app.get('/list', function(req, resp){
  runListGet(req, resp);
});

async function runListGet(req, resp) {
    try {
      const posts = db.collection(POSTS);
      const res = await posts.find().toArray();
      const query = { posts: res };
      resp.render('list.ejs', query);
    } catch (e) {
      console.error(e);
      resp.status(500).send('Error: could not read the list');
    }
}
```